

Root-Tweet Forward Prediction of Political Reply Dynamics

Technical Whitepaper
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Objective

This project tests whether downstream political discourse can be predicted from the root tweet alone. The target is distribution-level forecasting across political alignment, emotion, sentiment, and aggression, rather than exact text reconstruction.

System Summary

- Agent personas are built from historical Twitter data using five classifiers: political leaning, emotion, sentiment, hate, and offensive language.
- Each holdout simulation starts with only the root tweet in metadata (0-shot mode; no target-thread replies used as generation input).
- A Mesa ABM runs 10 synchronous rounds. Agent reply likelihood is behaviorally weighted by aggression.
- Validation reclassifies real and simulated tweets with the same five-model stack and scores JSD-based similarity plus aggression-mean alignment.

Forward Holdout

Target account: Sen. Marsha Blackburn (user_id 278145569).
Protocol: 8 chronological root tweets found; first 3 used for responder-profile construction; last 5 held out for strict forward prediction.
Per-run prediction input: root tweet text only (source=forward_holdout_zero_shot, use_few_shot=false); 200 synthetic agents sampled from 124 matched historical responders.

Holdout Root Tweet ID	Accuracy (%)
1800961685949198714	97.93
1800996791078236518	94.83
1801248635654390240	94.58
1801290362700394984	90.57
1801402597707767909	94.79
Mean ± SD	94.54 ± 2.34

Component means across the 5 predicted tweets: political similarity 96.45%, emotion similarity 91.78%, sentiment similarity 97.39%, aggression similarity 90.17%.

Interpretation and Limits

- Using only the root tweet and history-derived responder personas, the system predicted the next five held-out conversations with 94.54% mean weighted accuracy. This complements broader project evidence of 92.3% mean accuracy across 100 held-out threads.
- The 94.54% score reflects distributional fidelity, not exact tweet-by-tweet text matching.
 - Reply-volume calibration was not directly optimized by this metric and should be a next target.
 - External validity remains bounded to USC X 24 political discourse and tested model settings.

Bottom line: historical tweets from a specific political account can be transformed into a predictive behavioral population that anticipates likely reaction patterns to that account's next root tweets before they are observed.